Examples for non-ideal solutions showing negative deviation: Acetone + chloroform, Chloroform + diethyl ether, Acetone + aniline,Chloroform + Benzene.

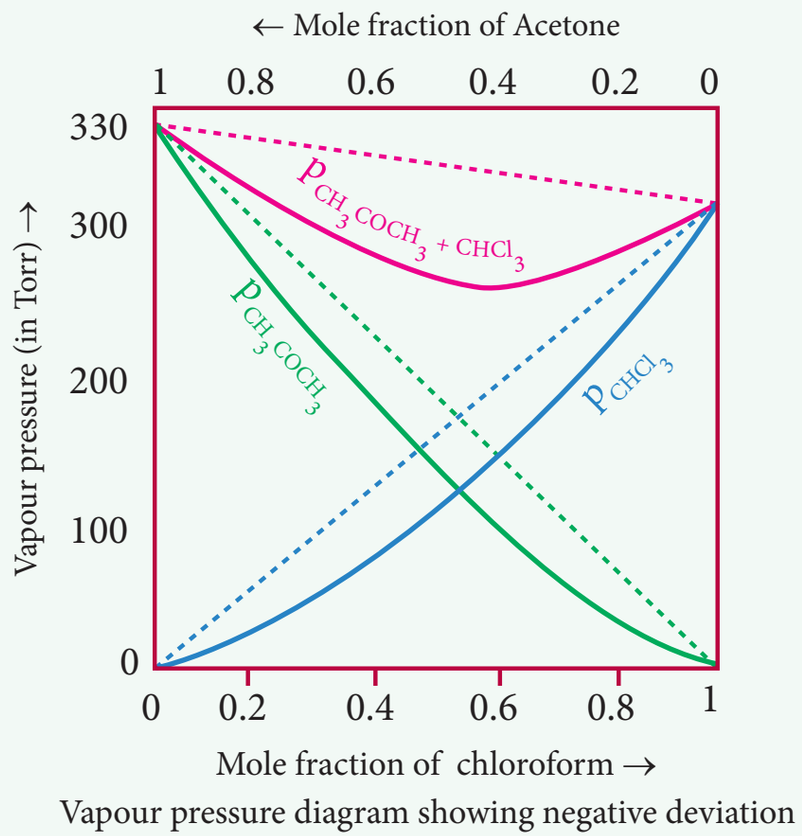

Figure 9.8 Negative deviation from Raoult's law. The dotted line (-----) is ideal behavior and the solid lines (____) is actual behaviour

9.8.3 Factors responsible for deviation from Raoult's law

The deviation of solution from ideal behavior is attributed to the following factors.

i) Solute-solvent interactions

For an ideal solution, the interaction between the solvent molecules (A-A),the solute molecules (B-B) and between the solvent & solute molecules (A-B) are expected to be similar. If these interactions are dissimilar, then there will be a deviation from ideal behavior.

ii) Dissociation of solute

When a solute present in a solution dissociates to give its constituent ions, the resultant ions interact strongly with the solvent and cause deviation from Raoult's law.

For example, a solution of potassium chloride in water deviates from ideal behavior because the solute dissociates to give K^+ and Cl^- ion which form strong ion-dipole interaction with water molecules.

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